OVERVIEW OF THE COLLECTION

CLAUDE OPUS 5.5, GPT-6-ASTRA, CLAUDE FABLE 5.1

Abstract. This collection contains two manuscripts, each resolving a published open question. Both main results are formally verified in Lean 4 with Mathlib. The manuscripts, Lean code and supporting computations were produced by AI models working as a team of agents; see the repository README for how the results were produced.

001. The maximum number of non-equivalent ancestral configurations

Discipline: mathematical biology (phylogenetics); enumerative combinatorics.

In the multispecies coalescent, gene tree probabilities are computed by dynamic programming over *ancestral configurations*. Disanto and Rosenberg (Bull. Math. Biol. 2019) introduced the number $c^*(T)$ of non-equivalent root configurations of a matching gene tree / species tree pair of shape T and conjectured that

$$\max_{|T|=n} c^*(T) = \left\lfloor 3^{\lfloor n/3 \rfloor}\, \tfrac{2r^2+8r+23}{27} \right\rfloor, \qquad r = n \bmod 3,\ n \geq 2,$$

having verified it for $n \leq 20$. The manuscript proves the conjecture for all n, shows that the maximizing shape is unique for $n \geq 9$ (it is obtained by repeatedly joining a three-leaf tree), determines the growth constant $3^{1/3}$ (previously known to lie in $[1.4422, 1.5028]$), and shows that the minimum is $n-1$, attained by exactly two shapes for $n \geq 4$. The key tool is the identity $s([L,R]) = 1 + s(L)s(R) - \binom{k}{2}$ for the number s of pruned shapes, with k the number of pruned shapes common to L and R. *Formalization:* two independent Lean proofs of the maximum, of uniqueness for $n \geq 9$ and of the minimum value.

002. The reversed-prime iteration of OEIS A100475 never enters a cycle

Discipline: elementary number theory.

OEIS A100475 iterates $k \mapsto \mathrm{rev}(p_k)$, where p_k is the k-th prime and rev reverses decimal digits, and asks whether a start other than 1 can lead to a loop; the question is an open statement in google-deepmind/formal-conjectures. The note shows that no start leads to a cycle: the number of digits never decreases along an orbit, $\pi(10^d) < 10^{d-1}$ for $d \geq 5$ confines any cycle to $[1, 1229]$, and a finite check excludes cycles there. In other bases cycles can exist: base 6 has the fixed point 11 and base 8 the fixed point 13. *Formalization:* a Lean proof of the negation of the formal-conjectures statement.

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